ENTRY 033 Confirming the Channel-Independence Ceiling — August 13, 2026 — Status: COMPLETE

Entry 033 — Every Formula Is Right, and the Route Still Loses 20 Points

bell_far_0_126's Regression Traces to the Same Structural Limit Entry 029 Found, Now Confirmed on an Unrelated Route

1. Isolating bell_far_0_126 on Sherbrooke

Entry 032 improved the reference set overall but made bell_far_0_126 and bell_mid_0_58 worse (gaps up to 20.39 points). Rather than guess, each channel was isolated against Aer the same way Entries 028 and 029 did for bell_mid_77_100 and bell_near_57_61 -- this route crosses 25 real SWAP legs (76 total two-qubit gate applications) toward qubit 126, the longest in the whole project.

channel	emulator predicts	Aer isolated	gap
gate error (Entry 032 formula)	84.51%	83.28%	1.23 pts
decoherence (Entry 028 T1 formula)	86.44%	89.72%	3.28 pts
readout error	95.91%	95.70%	0.21 pts

Every single formula checks out well individually -- the largest gap is 3.3 points, on decoherence. None of Entries 028's, 029's, or 032's fixes are wrong.

2. Where the Gap Actually Comes From

Multiplying the three Aer-ISOLATED results together (as if the channels were independent) gives 0.8328 × 0.8972 × 0.9570 = 71.5%. The REAL combined-noise Aer result for this circuit is 49.51%. A 22-point gap appears purely from treating three genuinely well-measured, individually-correct channels as if they don't interact -- with nothing left to blame on any of this project's formulas, since each one was just independently verified against its own isolated ground truth.

Entry 033 — Conclusion

This is not a new problem. It is Entry 029's finding -- "even a perfect decomposition of each channel in isolation, recombined by simple multiplication, misses a real interaction effect" -- showing up again on a completely different route, on a different chip, driven by a different cause (25 accumulated SWAP legs here, versus one catastrophically low-T1 qubit on Kyiv's bell_near_77_82). Two independent circuits now show the same signature: individually correct channels, multiplicatively combined, systematically overshoot real fidelity once a route accumulates enough total exposure -- whether that exposure comes from one extreme qubit or from many ordinary ones. This confirms it as a general property of the model's multiplicative-composition assumption, not a coincidence specific to qubit 80 on Kyiv.

3. Why This Isn't Chased Further Right Now

Fixing this for real requires modeling how gate error and decoherence correlate along a route -- effectively simulating correlated noise rather than composing independently-measured channel probabilities, which is close to what Aer itself already does at the circuit level. That is a materially different, larger undertaking than any fix shipped this session, and outside a closed form the rest of this project's formulas (Entries 025, 028, 032) were built around. The current model's real, verified achievement stands: mean gap on Kyiv's reference set fell from 13.28 (Entry 026) to 4.06 points (Entry 032), and on Sherbrooke from 9.90 to 8.66, with every individual physical channel -- gate error, decoherence, readout -- now independently verified correct. The two longest routes on each chip are a known, understood, documented ceiling on that approach, not an unexplained failure.

4. Where This Leaves the Project

The physics-debugging arc that ran from Entry 022 through this entry has taken v4.1 from a validation harness that silenced 53% of its own noise model to a closed-form emulator with no fitted, chip-specific coefficients, verified on two chips it wasn't tuned on. The remaining gap is well-characterized rather than mysterious. That is a reasonable, honest stopping point for this arc -- the next entry moves to building the batch-circuit dataset for the ML phase, using this validated model as the benchmark any learned model will need to beat.

Research Log — Index Addendum (Entry 033)

Entry	Title	Date	Status
Entry 033	Confirming the Channel-Independence Ceiling on a Second Route	Aug 13, 2026	Complete

No code changes. Confirms Entry 029's channel-independence finding generalizes beyond Kyiv's qubit 80 to any route with enough cumulative exposure. Closes the Entries 022-033 physics-debugging arc; next entry begins the ML batch-dataset phase.